

When Events Act: Topology-Aware Tracklet Filtering for Traffic Video Retrieval

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Abstract. Large-scale video retrieval commonly relies on sparse keyframes, which are effective for appearance-based queries but may miss motion-dependent events. This limitation is particularly important in traffic surveillance, where maneuvers such as turning or making a U-turn depend on vehicle motion relative to road structure rather than on a single frame. We propose a topology-aware tracklet filtering mechanism that augments keyframe-based semantic retrieval with structured traffic-motion evidence. Vehicle detections are associated into tracklets, interpreted with respect to inferred road topology, and converted into traffic-action sequences with temporal intervals. These intervals are then used to filter or rerank semantic retrieval results without independently indexing every frame. On 89 human-verified traffic-maneuver queries, the proposed method improves the median target rank from 12,804 to 3,932, with 86 of 89 queries improved; for turning queries, the median rank improves from 13,068 to 3,013. The complete system additionally achieved a qualification score of 81.15 out of 86 in the Ho Chi Minh City AI Challenge 2026.

Keywords: Video retrieval · Object tracking · Trajectory reasoning · Temporal retrieval · Composed image retrieval · Interactive video search

1 Introduction

Large-scale video retrieval systems often represent videos using sparse keyframes and combine vision–language similarity with complementary signals such as OCR, ASR, object filtering, and temporal search [2, 7, 6]. While this representation is efficient, it can miss short-lived or motion-dependent events [11, 12], whereas increasing frame density introduces substantial redundancy.

This problem is particularly evident in traffic surveillance. Maneuvers such as turning left, turning right, going straight, or making a U-turn depend on how a vehicle moves relative to the surrounding road structure. The same maneuver may appear differently across camera viewpoints, while similar image-space

trajectories may correspond to different actions under different intersection layouts. This motivates reasoning jointly over vehicle trajectories and structured road context, as also reflected in map- and lane-aware motion modeling [10, 5]. Therefore, motion reasoning based only on individual frames or raw displacement is often insufficient.

We address this limitation with a **topology-aware tracklet filtering** mechanism. Consecutive vehicle observations are associated into tracklets and interpreted relative to inferred road topology to produce structured traffic actions and their temporal intervals. These cues are used to filter or rerank the results of the keyframe-based semantic retriever, adding motion-sensitive evidence without treating densely sampled frames as independent retrieval units. The proposed module extends our previous WESp framework [11], where event-aware temporal reranking remains responsible for relationships across multiple events.

The main contributions of this work are: (1) a topology-aware representation that converts vehicle tracklets into structured traffic-action sequences; (2) a tracklet-based filtering and reranking mechanism that injects temporal motion evidence into keyframe retrieval; and (3) an evaluation on 89 human-verified traffic-maneuver queries, where the proposed method improves the median target rank from 12,804 to 3,932, including an approximately $8\times$ median rank gain for turning queries.

The complete multimodal retrieval system achieved a qualification score of 81.15 out of 86 across the KIS, VQA, and TRAKE tasks of the *Ho Chi Minh City AI Challenge 2026*.

2 Related Work

2.1 Keyframe-Based and Temporal Video Retrieval

Interactive video retrieval systems commonly represent large video collections through compact sets of keyframes and combine vision-language similarity with complementary modalities such as OCR, ASR, object filtering, and query refinement. Recent VBS systems continue to follow this multimodal paradigm while improving scalability and interaction efficiency. Fusionista2.0, for example, combines multiple retrieval modalities with efficient preprocessing and interactive reranking [9], while the latest NII-UIT system extends multimodal retrieval with temporal hotspot localization and dedicated in-video search for fine-grained evidence discovery [15].

A growing body of work has also moved beyond independent keyframe matching. NII-UIT previously introduced Dynamic Temporal Search to restrict subsequent queries to temporal neighborhoods around retrieved candidates [7]. WESp combines MLLM-based event capturing with temporal reranking to recover short-lived events and maintain coherence across event sequences [11], while Poly-Temporal Search jointly models compositional and sequential constraints [12]. More recently, H-EAGLE organizes video content hierarchically at image, shot, and action levels [13], and Exquisitor revisits temporal querying

by constructing ordered chains from multiple query stages [8]. These approaches provide increasingly rich representations of temporal context, but they primarily reason over frames, segments, shots, or semantic events. They do not explicitly interpret the trajectory of an individual vehicle relative to the topology of a traffic scene.

2.2 Vehicle Tracking and Topology-Aware Traffic Understanding

Multi-object tracking associates detections across consecutive frames to recover persistent object trajectories. Tracking-by-detection methods such as SORT, DeepSORT, ByteTrack, and BoT-SORT improve temporal association through motion, appearance, or confidence-aware matching [3, 18, 19, 1]. More recent trackers continue to improve robustness under occlusion and crowded scenes [14]. However, these methods primarily aim to maintain object identity; the resulting tracklets do not directly specify the semantic traffic maneuver performed by a vehicle.

For road scenes, trajectory semantics are closely related to the surrounding road geometry. Map- and lane-aware motion models therefore represent lane connectivity and local road structure together with object trajectories rather than relying only on image-space displacement [5, 10]. This observation is particularly relevant to traffic video retrieval, where actions such as turning left, turning right, going straight, or making a U-turn are defined relative to the entry and exit structure of an intersection. Our work connects these two lines of research by converting vehicle tracklets into topology-aware traffic actions and using their temporal intervals as structured filters for keyframe-based retrieval.

3 Multimodal Video Retrieval System

3.1 Retrieval Backbone and Multimodal Filtering

Our system builds on the event-aware retrieval framework introduced in WESp [11]. Videos are represented by representative keyframes, which form the primary retrieval index. SigLIP2 [16] and BEiT-3 [17] provide complementary image-text representations, and their ranked outputs are combined to obtain the initial semantic candidates.

The semantic retriever is complemented by modality-specific evidence from OCR, ASR, object tags, and event descriptions. These signals are primarily used as optional filters rather than mandatory components of the retrieval score, allowing each query to activate only the evidence it requires. Event descriptions extracted from short video segments provide additional temporal context, while the temporal reranking mechanism inherited from WESp is retained for queries involving multiple ordered events.

For speech-oriented queries, we use Qwen3-ASR to generate Vietnamese transcripts over overlapping 20-second audio windows. Audio is normalized to 16-kHz mono and transcribed with a 2-second overlap to preserve speech near segment boundaries. We further normalize spoken numerical expressions using a

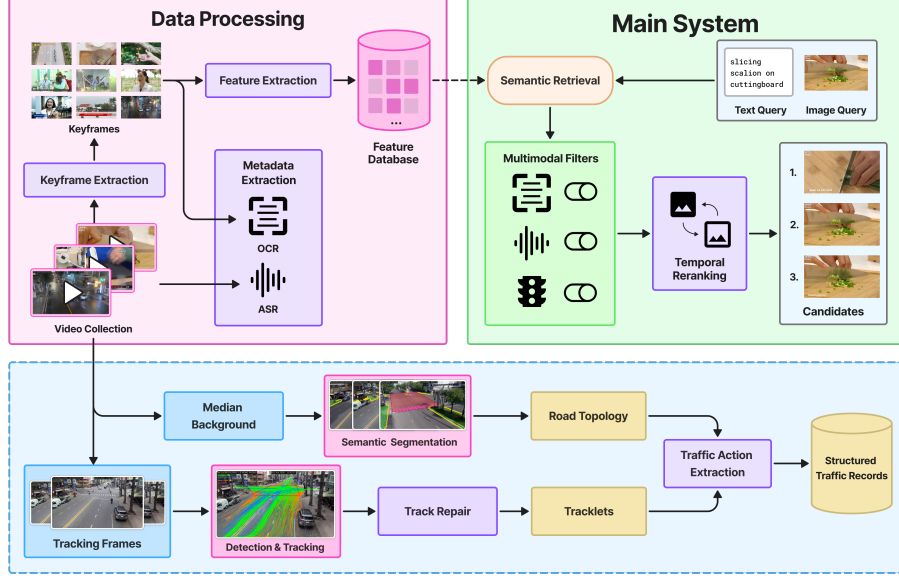


Fig. 1. Overview of the retrieval system. Keyframe-based semantic search remains the primary retrieval backbone. AzaleaOCR provides a dedicated searchable scene-text layer, while ASR, object tags, event descriptions, composed-image refinement, and topology-aware tracklets provide complementary evidence when required by the query.

Qwen3.5-based post-processing stage, producing the canonical ASR corpus of 55,818 timestamped segments across 1,487 videos. These transcripts provide an auxiliary lexical retrieval signal, supporting accented and accent-insensitive keyword or fuzzy filtering alongside the primary visual search pipeline.

For text-oriented queries, we use *AzaleaOCR*, combining DeepSolo++ localization with PARSeq and our Vietnamese-adapted SVTRv2-S recognizer. SVTRv2-S-VI-Scene is fine-tuned from Union14M-L-Filter using a two-stage mix of VinText crops, BKAI2023 data, and 80,000 synthetic Vietnamese samples designed around diacritics and scene-text degradations. A 228-character NFC-aware vocabulary preserves Vietnamese orthography. At inference, SVTR handles confident predictions while uncertain cases fall back to PARSeq. The resulting system achieved 0.652 word-level end-to-end H-mean on held-out SignboardText and processed 274,488 keyframes from 873 AI Challenge 2026 videos. Recognized text is indexed as an auxiliary lexical signal for retrieval.

The current system further extends this backbone with two additional capabilities. Composed image retrieval supports interactive refinement from a selected keyframe, while a topology-aware tracklet filter provides structured traffic-motion evidence. Both modules operate on top of the original semantic retrieval pipeline rather than replacing it.

3.2 Composed Image Retrieval

When a visually relevant keyframe has already been identified, the system supports composed image retrieval (CIR) to refine the result while preserving its visual context. Let $r = f_{\text{img}}(I_{\text{ref}})$ denote the normalized embedding of a reference keyframe, and let a and m denote the text embeddings of concepts to add and remove. The composed query is defined as

$$d = \text{norm}(a - m), \quad q_{\text{cir}} = \text{norm}(r + sd), \quad (1)$$

where s controls the strength of the modification. For addition-only and removal-only queries, d reduces to a and $-m$, respectively.

Because direct nearest-neighbor retrieval may favor near-duplicates of the reference image, a broad candidate set is reranked using composed-query similarity, target-concept agreement, preservation of the original context, and consistency with the requested semantic direction. CIR is used as an interactive refinement tool and is independent of the traffic-specific tracklet branch.

3.3 Interactive Retrieval and Evidence Inspection

To facilitate rapid candidate verification and query refinement, the platform provides an interactive web-based retrieval interface optimized for human-in-the-loop video search. The system seamlessly unifies multimodal query formulation, dynamic metadata filtering, and synchronized evidence inspection within a unified environment.

Multimodal Evidence Inspection and Temporal Verification. To accelerate result validation, the system overlays object motion trajectories directly onto candidate keyframes, visually rendering continuous vehicle tracklets. Extracted OCR text and ASR transcripts are presented alongside keyframes and maintained in real-time synchronization with video playback. The interface supports in-transcript keyword filtering and automated timestamp seeking upon segment selection, enabling precise temporal target verification.

Temporal Browsing and CIR Refinement. The interface features an expanded keyframe viewer that supports continuous browsing through contiguous neighboring shots across the entire video duration. Furthermore, the platform integrates a Composed Image Retrieval (CIR) module, enabling text-guided visual modifications (e.g., adding, altering, or removing visual objects/attributes) relative to a reference keyframe to dynamically re-query the backend.

Dynamic Filtering and Keyboard Navigation. Auxiliary metadata filters—including OCR keyword matching, ASR transcripts, video constraints, and traffic-specific structured constraints—dynamically execute backend re-queries upon update. To maximize operational throughput during competitive search tasks, the system incorporates comprehensive keyboard shortcuts across all navigation, filtering, and submission workflows.

Traffic-Specific Structured Filtering. For traffic-oriented queries, the interface exposes the structured constraints produced by the topology-aware



Fig. 2. Example semantic retrieval interface of the proposed system. For the query “red bus,” visually relevant keyframes are ranked according to the image–text retrieval backbone and presented in an interactive grid for rapid candidate inspection and further refinement.

tracklet branch directly to the user. Vehicle type and maneuver labels, including *stop*, *go straight*, *turn left*, *turn right*, and *U-turn*, can be selected independently or combined into an ordered action sequence. The interface can either promote matching candidates within the semantic ranking or restrict the result set to tracklets satisfying the selected constraints. This design allows users to combine free-form semantic retrieval with explicit motion evidence when the target event cannot be reliably identified from a single keyframe.

4 Topology-Aware Tracklet Filtering for Traffic Video

4.1 Vehicle Tracklet Construction

Traffic videos are processed at a denser frame rate than the keyframe retrieval index. A vehicle detector and multi-object tracker associate consecutive detections into tracklets. For geometric reasoning, each detection is represented by the mid-point of the bottom edge of its bounding box, which provides a more meaningful approximation of the vehicle position on the road than the box center.

Only vehicle classes relevant to the traffic-action vocabulary are retained. The final vehicle type is obtained from the majority detector label over the track. Association is deliberately conservative because an incorrect identity merge can create false motion patterns. After online tracking, fragmented tracklets are re-connected only when their location, direction, vehicle type, and scale are suffi-

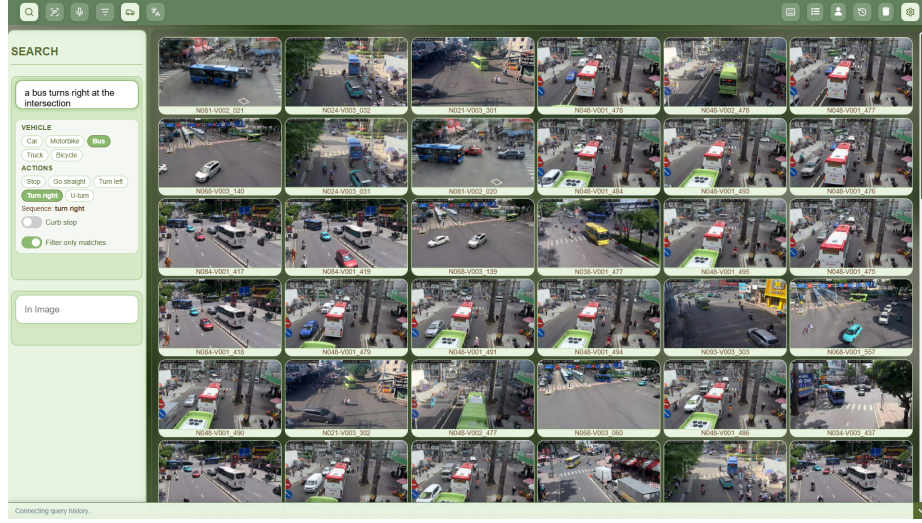


Fig. 3. Traffic-oriented retrieval interface with structured tracklet filtering. For the query “a bus turns right at the intersection,” the user combines free-form semantic search with explicit vehicle and maneuver constraints. The interface supports vehicle-type filtering, traffic-action selection, ordered action sequences, and optional match-only filtering, allowing topology-aware motion evidence to refine or restrict the retrieved candidates.

ciently consistent; ambiguous associations remain separated. Tracks that disappear while the vehicle is still moving are marked as *truncated* for later handling.

4.2 Topology-Aware Traffic Action Extraction

Each validated tracklet is converted into an ordered sequence of five traffic actions: *stop*, *go straight*, *turn left*, *turn right*, and *U-turn*. Before action extraction, the ground-contact trajectory is smoothed to reduce detector jitter.

A *stop* is assigned only when the vehicle remains spatially stable for a sufficient duration, allowing slow traffic movement to be distinguished from a genuine stop. The *go straight* label is treated as a traffic decision rather than a persistent motion state: it is emitted when a vehicle proceeds through the scene without another maneuver or resumes straight movement after a stop.

Directional maneuvers are interpreted primarily from the relation between the trajectory and the local road topology rather than from image-plane turning angles. A camera-specific scene map is automatically inferred from the static background to identify road and intersection structure. For a vehicle crossing an intersection, its entry and exit boundaries determine whether the maneuver corresponds to left, right, or straight movement. Image-space direction changes are used only as consistency checks. U-turns are handled more conservatively

and require a near reversal with sufficiently consistent incoming and outgoing trajectory segments. When the scene topology or trajectory is unreliable, the system avoids assigning a directional action rather than forcing a potentially incorrect label.

4.3 Structured Tracklet Filtering

The tracklet branch acts as a structured filter rather than an independent retrieval index. A traffic constraint may specify a vehicle type, an ordered action sequence, and optional scene-related attributes such as curb stopping. For tracklet $\mathcal{T}_i$, let $A_i = [a_{i,1}, \dots, a_{i,m}]$ denote its ordered action sequence, with $I_{i,j}$ representing the temporal interval of action $a_{i,j}$.

A requested action sequence is matched as a contiguous subsequence of A_i . For example, *stop* $\rightarrow$ *turn left* can match *go straight* $\rightarrow$ *stop* $\rightarrow$ *turn left*. Only the temporal intervals corresponding to the matched actions are used to modify the keyframe ranking, rather than the complete lifetime of the vehicle track.

For a candidate keyframe f at timestamp t_f , let $B(f, q_{\text{trk}})$ indicate whether the frame overlaps an action interval from a tracklet satisfying the requested structured constraints. The final score is

$$S(f \mid q) = S_{\text{sem}}(f, q_{\text{sem}}) + \lambda B(f, q_{\text{trk}}), \quad (2)$$

where λ controls the contribution of tracklet evidence. Without a traffic constraint, $B = 0$ and the system reduces to the standard semantic retriever.

For multi-event queries, tracklet filtering is applied to the relevant individual event before the existing temporal reranking stage. The proposed branch therefore provides local maneuver-specific evidence, while WESp continues to model chronological consistency across multiple events [11].

5 Experiments

5.1 Dataset

We evaluate the system on the video collection of the Ho Chi Minh City AI Challenge 2026 (HCMAI26) [4], which is organized into four subsets: L, M, N, and S. The corpus contains 1,487 videos with a total duration of about 325.96 hours and covers diverse content, including news broadcasts, cultural and educational programs, fixed-camera traffic surveillance (CCTV), and cycling tournaments (Table 1). Video length varies widely, from 16 seconds to over 5.6 hours. From these videos we obtain 1,060,556 keyframes in total.

The traffic-specific tracklet branch is primarily applied to fixed-camera CCTV footage in Subset N, where road topology can be inferred consistently across the video. Cycling footage is included in the broader retrieval corpus but is not used for topology-aware maneuver analysis because of its moving viewpoint.

Table 1. Statistics of the evaluation corpus.

Subset	Content	Videos	Dur. (h)	Keyframes
L	Multi-category	873	130.68	187,347
M	News	304	99.55	535,925
N	Traffic CCTV	298	49.14	176,359
S	Cycling	12	46.60	160,925
Total		1,487	325.96	1,060,556

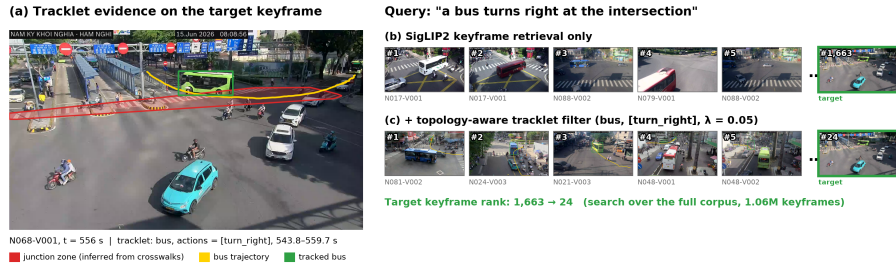


Fig. 4. Qualitative example of topology-aware traffic filtering. (a) The target keyframe and its supporting tracklet evidence, where the tracked bus is associated with a right-turn action. (b) SigLIP2 keyframe retrieval alone ranks the correct result at position 1,663. (c) Applying the structured traffic constraint (*bus*, *turn_right*) promotes the same target to rank 24 over the full corpus of approximately 1.06 million keyframes.

5.2 Qualitative Traffic Retrieval Analysis

Figure 4 illustrates the effect of the proposed tracklet filter on a motion-dependent traffic query. The query asks for “a bus turns right at the intersection.” With SigLIP2 keyframe retrieval alone, the correct target is ranked at position 1,663 over the full corpus. After applying the structured traffic constraint (*bus*, *turn_right*), the same target is promoted to rank 24.

The remaining ranking gap is expected because the current tracklet filter uses only coarse structured traffic attributes, in this example the vehicle type and maneuver label. It does not yet incorporate richer scene or event descriptions that could further distinguish visually similar traffic moments. Nevertheless, the example shows that trajectory-derived evidence can substantially narrow the search space when the requested action is not reliably expressed by a single keyframe.

5.3 Qualification Round Result

In the qualification round of the *Ho Chi Minh City AI Challenge 2026*, our system achieved a score of **81.15 out of 86**, corresponding to **94.36%**, across the KIS, VQA, and TRAKE tasks. This result reflects the performance of the

complete retrieval system, including multimodal filtering, temporal reranking, and interactive retrieval components.

5.4 Quantitative Evaluation of Tracklet Filtering

We further evaluate the traffic-specific branch on 89 human-verified maneuver queries from Subset N, comprising 60 turning queries and 29 straight-motion queries. For each query, the baseline ranks all 176,359 CCTV keyframes using SigLIP2 text-image similarity. We compare this baseline with two uses of the proposed tracklet evidence: hard filtering, which retains only keyframes overlapping a matching vehicle-action interval, and score boosting according to Eq. 2 with $\lambda = 0.05$. A retrieval is considered correct when the ranked keyframe falls within the temporal span of the target vehicle.

Table 2. Traffic-maneuver retrieval with and without topology-aware tracklet evidence. Median rank is lower while MRR is higher. Improved/worse reports the number of queries whose target rank improves or degrades relative to the SigLIP2 baseline. Median gain is the median ratio between the baseline rank and the resulting rank.

Queries	Method	Med. rank ↓	MRR ↑	Imp./worse	Gain
Turns (60)	SigLIP2	13,068	0.0002	–	–
	+ tracklet filter	3,013	0.0016	57 / 3	×8.0
	+ tracklet boost	3,035	0.0016	57 / 3	×7.7
Straight (29)	SigLIP2	9,961	0.0037	–	–
	+ tracklet filter	9,018	0.0040	29 / 0	×1.1
	+ tracklet boost	9,018	0.0040	29 / 0	×1.1
All (89)	SigLIP2	12,804	0.0013	–	–
	+ tracklet filter	3,932	0.0024	86 / 3	×5.1
	+ tracklet boost	4,040	0.0024	86 / 3	×5.1

Topology-aware tracklet evidence substantially improves retrieval for directional maneuvers. For the 60 turning queries, median target rank decreases from 13,068 to approximately 3,000, with 57 of 60 queries improving and a median rank gain of about 8×. Hard filtering and score boosting produce nearly identical results, suggesting that the benefit primarily comes from identifying the relevant maneuver intervals rather than from the particular fusion strategy. In contrast, *go straight* provides only limited discrimination, since straight motion is common among vehicles in traffic scenes, resulting in a much smaller gain of approximately 1.1×.

6 Discussion and Future Work

The current system demonstrates that trajectory evidence can be integrated effectively into a multimodal video retrieval pipeline without replacing its keyframe-

centered design. The system achieved **81.15 out of 86 questions correctly** in the qualification round of the *Ho Chi Minh City AI Challenge 2026*, while supporting semantic retrieval, OCR and ASR filtering, temporal reranking, composed-image refinement, and topology-aware traffic reasoning. These components serve complementary roles: semantic and multimodal retrieval identify visually and contextually relevant candidates, while the trajectory branch provides additional evidence when the meaning of an event depends on object motion.

The main advantage of the current trajectory branch is its ability to convert continuous vehicle motion into structured traffic actions and temporal intervals. This allows the system to distinguish events such as stopping, turning left, turning right, going straight, and making a U-turn even when a single keyframe provides insufficient temporal evidence. Importantly, the tracklet branch is used as a structured filter on top of the existing retrieval backbone, so trajectory processing is activated only when motion-related evidence is relevant to the query. This preserves the efficiency of the keyframe-based representation while adding object-level temporal information.

A remaining limitation is that the current tracking stage primarily provides **trajectory-level information rather than reliable vehicle-level identity**. ByteTrack can associate detections into continuous paths, but in dense traffic or under occlusion, the association may still be fragmented or switched between visually similar vehicles. As a result, the system can determine that a particular trajectory corresponds to a left turn or a stop, but it may not always reliably determine *which specific vehicle* generated that trajectory. This limitation becomes particularly important when several vehicles perform similar maneuvers simultaneously.

Future work will therefore focus on strengthening **vehicle-specific trajectory association**. In particular, we plan to improve the current ByteTrack-based tracking stage with stronger appearance and identity cues so that each trajectory can be consistently associated with the same physical vehicle throughout the scene. This would allow the retrieval system to reason not only about *what trajectory occurred*, but also *which vehicle performed it*, enabling more precise queries involving vehicle-specific actions, interactions, and sequential behavior. Further improvements can also address identity switches and fragmented tracks under occlusion, while retaining the current topology-aware action extraction and keyframe-based retrieval architecture.

7 Conclusion

We introduced a topology-aware tracklet filtering mechanism for motion-dependent traffic video retrieval. The proposed method converts vehicle tracklets into structured traffic actions with temporal intervals and uses them as auxiliary evidence to filter or rerank keyframe-based semantic retrieval results. This allows the system to capture motion-dependent events without requiring dense frame indexing.

On 89 human-verified traffic-maneuver queries, the proposed method improves the median target rank from 12,804 to 3,932, with 86 of 89 queries im-

proved. For turning maneuvers, the median rank decreases from 13,068 to 3,013, corresponding to an approximately $8\times$ median rank gain. The complete multi-modal retrieval system additionally achieved a qualification score of 81.15 out of 86 in the Ho Chi Minh City AI Challenge 2026.

The current approach remains limited by fragmented tracks and identity switches in dense traffic, as well as by a relatively coarse traffic-action vocabulary. Future work will focus on stronger vehicle association and richer motion and interaction modeling for more fine-grained traffic retrieval.

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